TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Resistance of a Wire by Post Office (P.O.) Box

The Wheatstone bridge packed into one box: two ratio arms chosen by plugs, one adjustable arm, and the unknown completing the diamond — balance means no current through the galvanometer.

PHY-1204 · PHYSICS-II SESSIONAL · EXPERIMENT 4 · LEVEL 1 TERM 2

1. Aim and theory

Aim: to measure the resistance of a given wire with a Post Office box. The P.O. box (named for its use by telegraph engineers of the Post Office) is a self-contained Wheatstone bridge. Four resistances form a diamond: the **ratio arms** P and Q, each selectable as 10, 100 or 1,000 Ω by plugs; the **adjustable arm** R, a full resistance box from 0.1 Ω upward; and the **unknown** X connected to the fourth pair of terminals. The battery sits across one diagonal of the diamond and the galvanometer across the other. When the galvanometer shows no deflection on pressing the keys, the bridge is balanced and the Wheatstone condition holds:

$$P / Q = R / X \quad \rightarrow \quad X = R \times (Q / P)$$

The ratio Q/P (the **multiplier**: ×0.01, ×0.1, ×1, ×10, ×100) is chosen so that R falls in a comfortable mid-range of its box, where its steps are finest relative to the value. Because plugs step R in 0.1 Ω jumps, the exact null usually lies between two plugs; it is pinned down by noting the galvanometer deflections on either side and interpolating.

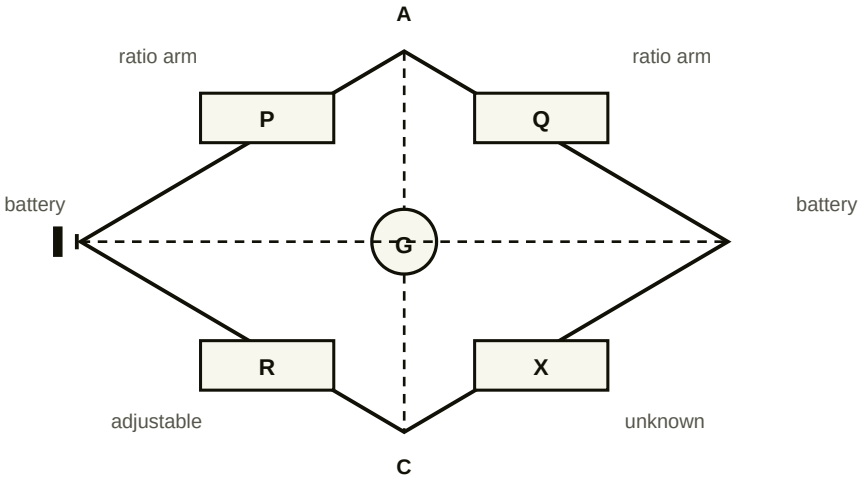


Figure 1 — The Wheatstone diamond inside the P.O. box. Battery across the A–C diagonal, galvanometer across the other; at balance the galvanometer diagonal carries no current and P/Q = R/X.

2. Apparatus and procedure

- **Apparatus:** P.O. box, the wire whose resistance is required (mounted between terminals or as a coil), galvanometer, cell, connecting wires.
- Connect the unknown across the X terminals, the cell and galvanometer to their marked terminals. Tighten every plug not in use — a loose plug adds its contact resistance to an arm.
- Set the ratio arms equal (P = Q = 10) for a first estimate: vary R for the roughest balance (galvanometer nearly zero). This tells the order of X.
- Choose the multiplier so R will be large (hundreds of ohms if possible): raise the ratio step by step, rebalancing each time, until R is as large as the box allows while the deflection still responds crisply to a 0.1 Ω change.
- Bracket the null: record the deflections at the two R values straddling balance and interpolate the exact R_0.
- Reverse the battery (or interchange arms) and repeat; average. Compute $X = R_0 \times (Q/P)$.

3. Worked calculation

Ratio arms set P = 100 Ω, Q = 10 Ω, so the multiplier is Q/P = 0.1. Balance is straddled at R = 47 Ω (deflection 3 divisions to the right) and R = 48 Ω (2 divisions to the left). Interpolating, the null sits at $R_0 = 47 + 3/(3 + 2) = 47.6$ Ω.

$$X = 47.6 \times 0.1 = 4.76\ \Omega$$

Had the ratio been left at ×1, R would have balanced near 4.7 Ω, where a 0.1 Ω plug step is a 2 % jump; at ×0.1 with R near 48 Ω the same step is 0.2 % — the multiplier is bought precision.

4. Precautions

- Press the battery key *before* the galvanometer key, and release in the reverse order, so self-induction kicks never reach the galvanometer unprotected.
- Plugs must be twisted in firmly; a high-resistance plug contact falsifies its arm.
- Keep currents small and momentary — the unknown wire and the box coils warm up and drift.
- Check the box's zero: with X shorted by a thick copper strip, the reading should be ~0 (any residual is lead/contact resistance to subtract).

Practice (viva preparation)

1. State the Wheatstone balance condition and why the galvanometer current is zero at balance.
2. With P = 10, Q = 1,000 and balance at R = 23.4 Ω, find X. Is this a good multiplier choice for a small unknown? Comment.
3. Why is the unknown not simply measured with an ohmmeter-style ammeter–voltmeter pair at this accuracy?
4. Deflections are +4 div at R = 62 Ω and −1 div at R = 63 Ω. Interpolate R_0.
5. Why must unused plugs be kept tight in a P.O. box?

Answer key

1. At balance, P/Q = R/X. The two ends of the galvanometer diagonal then sit at the same potential (the drops along the two sides match), so no current flows through it.
2. X = 23.4 × (1000/10) = 2,340 Ω. The multiplier ×100 with R only 23.4 Ω wastes the box: a 0.1 Ω step is 0.4 % of R. Better to drop the multiplier so R climbs into the hundreds.
3. The ammeter–voltmeter method loads the circuit and is limited by meter class accuracy (~1–2 %); the bridge is a null method — the galvanometer only detects zero, its calibration irrelevant, and accuracy rests on the standard resistances (~0.1 %).
4. $R_0 = 62 + 4/(4 + 1) = 62.8$ Ω.
5. Every plug in an arm is part of that arm's resistance; a loose plug inserts an unknown contact resistance in series, silently changing P, Q or R from its marked value.
